# Krish A. Patel

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## Objective

Electrical engineering student at Georgia Tech with experience in hardware design, scientific computing and optimization research, and a strong analytical foundation in energy systems, seeking an internship for summer 2027.

## Education

**Georgia Institute of Technology | Atlanta, GA** August *2026 – Present*
Bachelor of Science in Electrical Engineering Expected Graduation, May 2028

**Emory University | Atlanta, GA** *August 2023 – May 2026*
Dual Degree, Bachelor of Science in Applied Math and Physics, GPA 4.0
2026 Trevor Evans Award for Outstanding Undergraduate Achievement in Mathematics

## Skills

**Programming Languages:** C++, MATLAB, Python, Julia, Java
**Hardware:** PCB Design (KiCad), LTspice, Arduino, Teensy, Verilog, circuit prototyping and testing, soldering, oscilloscopes
**Scientific Computing:** CUDA, Numerical Optimization, Finite Element Methods (FEM)
**Tools:** Git, GitHub, LaTeX, Excel
**Professional Organizations:** Georgia Tech IEEE
**Communication:** Technical and research reports, design proposals, presentations (large and small audiences)
**Languages:** English (fluent), Hindi (fluent), Gujarati (fluent), German (beginner)

## Experience

**NSF REU in Model Aware and Data Driven Inference | Atlanta, GA** *May 2025 – August 2025*
***Emory University, Department of Mathematics***

- Solved for theoretically optimal low-rank linear models, in Bayesian risk sense, for scientific machine learning tasks such as autoencoding, forward surrogate models and inverse problems.
- Performed data analysis comparing optimal low-rank models against deep neural networks, using Python/PyTorch, on financial factor analysis and image reconstruction tasks; demonstrated that the optimal linear models match or outperform learned nonlinear architectures on multiple datasets. Published research in American Institute of Applied Mathematics journal.

## Projects

**Designing a custom PCB for an autoclave system** *August 2026 – Present*
**Undergraduate Researcher | Plasma and Dielectrics Lab, Georgia Tech**

- Designing and prototyping a custom PCB using KiCad, integrating sensor and relay interfaces to monitor and control real-time temperature and pressure in an autoclave system.
- Programing an embedded system on a Teensy microcontroller to process sensor data, drive relay switching logic, and transmit data wirelessly for real-time monitoring.
- Ensuring tighter thermal and pressure tolerances in the autoclave to better cure an epoxy insulation shell for a coaxial transformer, being developed as part of a modular power converter system.

**Computational PDEs and Optimization** *August 2025 – May 2026*
**Undergraduate Honors Thesis | Applied Math**

- Programmed numerical models in Python to simulate wildfire spread using nonlinear diffusion-advection-reaction partial differential equations.
- Applied Bayesian optimization techniques to identify optimal intervention strategies, minimizing expected damage from wildfires by 50% across varying environmental conditions.
- Presented findings to a committee and defended the undergraduate thesis project with Highest Honors.

## Relevant Coursework

**Electrical Energy Systems:** Principles of power systems and circuit analysis; AC and 3 phase power; electromechanical systems.
**Applied Electromagnetics:** Electrostatics and magnetostatics; transmission lines; time-varying fields and EM waves.
**Engineering Software Design:** Object oriented software methods for engineering applications (C++).